# PakEmis Sector Technical Note TN04

# Three Categories Partitioning One Waste Stream: Solid-Waste Disposal, Open Burning and Domestic Wastewater Emissions for Pakistan

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**Abstract**

Pakistan's waste sector has no current national greenhouse-gas estimate built from Pakistani activity data, and the international products that fill the gap disagree with each other by amounts larger than the mitigation decisions they inform. We report the sector for FY2023–24 at 37.45 Mt $CO_2$e (AR5, 36.37 Mt on AR6) across three IPCC categories: domestic wastewater (4D1) at 24.46 Mt, solid-waste disposal (4A) at 9.17 Mt and open burning (4C2) at 3.83 Mt. The current 10,000-draw conditional parameter ensemble gives 25.4 Mt to 61.5 Mt $CO_2$e (2.5th–97.5th percentiles). It preserves the asymmetric effluent-$N_2O$ factor treatment and the bounded waste-fate calculation; management alternatives and missing sources remain outside that interval. The same fate realisations support open-burning air pollutants, including 63.85 kt primary $PM_{2.5}$, without double-counting generated material. The national annual core extends the disposal-history state and supported population inputs through FY2024–25; this note retains its detailed FY2023–24 analysis.

Two features distinguish the estimate from a conventional compilation. First, solid-waste disposal and open burning are computed from one mass-balanced national waste-generation figure rather than from separate literature: 31.56 Mt generated, 50.3 % collected to disposal and the remainder uncollected, of which 60 % burns and the unburned 6.27 Mt is carried as an explicit unresolved fate with no emissions pathway applied. Disposed plus uncollected equals generated, asserted with exact equality rather than reconciled afterwards, so an argument about the collection rate moves emissions between two categories without creating or destroying waste. Inventories that estimate landfilling and open burning independently have no such constraint, and nothing then forces their sum to be a quantity of waste the country actually produces.

Second, the largest category rests on an assumption the guidelines leave genuinely open for Pakistani conditions. IPCC gives a methane correction factor of 0.5 for stagnant sewers and 0.1 for discharge to a flowing river; Pakistani urban drainage has properties of both. At 0.5 domestic wastewater is 24.46 Mt and the sector sits at 141 % of EDGAR's year-matched Pakistani waste total; at 0.1 it is 15.39 Mt and the sector lands close to EDGAR. The physical description of the drains supports the higher value and the only independent comparator supports the lower. We report both, adopt 0.5 as central because the physical description is the stronger evidence, and identify the measurement that would settle it: dissolved-methane sampling on urban drains across a seasonal cycle. The physical description and the external inventory comparison play different roles; agreement with EDGAR is not used to select the central parameter. Seasonal measurements of drain methane could discriminate between the retained structural interpretations.

*Keywords:* municipal solid waste, first order decay, open burning, domestic wastewater, methane correction factor, exhaustive partition, IPCC 2006 Guidelines, Monte-Carlo uncertainty, Pakistan, MRV

**PakEmis Sector Technical Note TN04.** Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

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**Release provenance.** This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

| | |
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| Release version | `1.0.0rc69` |
| Commit | `d8d874622f9a` |
| Reporting period | July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30) |
| Reference | S_TAN (household survey; TAN nitrogen) |
| FY2023–24 total | 444.61 Mt $CO_2$e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN |
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## 1. Introduction

Waste is the category of a national greenhouse-gas inventory where the arithmetic is easiest and the activity data hardest. The emission factors are few and well characterised; what is difficult is knowing how much waste a country generates, what fraction of it reaches a disposal site, what happens to the fraction that does not, and how the sewage of a large population is actually handled. None of those quantities is metered anywhere in Pakistan.

That difficulty has a specific consequence for how the sector is usually compiled. Because the activity data must be modelled rather than read, each category tends to be estimated on its own terms: landfill methane from a tonnage assumption, open burning from a per-capita burning rate in the literature, wastewater from a population and a default organic load. The three estimates are then reported together. Nothing in that procedure requires the waste sent to landfill and the waste openly burned to sum to a quantity of waste the country plausibly produces, and in general they do not.

### *1.1. What this paper does*

We report Pakistan's waste sector for FY2023–24 from Pakistani activity data, with the three categories constructed so that the constraint the conventional procedure lacks is enforced by construction. A single national waste-generation figure is partitioned: what is collected goes to the first-order-decay model for solid-waste disposal, what is not collected is what can burn, and the two sum to the generated quantity exactly. The partition is asserted in code with equality rather than checked afterwards with a tolerance, because a tolerance permits precisely the drift the constraint exists to remove.

The result is a sector total of 37.45 Mt $CO_2$e, of which domestic wastewater is two-thirds. It is the second most uncertain sector in the national inventory that contains it, and this paper is as much about where that uncertainty lives and what would reduce it as about the central estimate.

### *1.2. Why Pakistan, and why this sector now*

Pakistan is the fifth most populous country in the world, urbanising rapidly, and its most recent full national inventory submission reports for FY2017–18 (Ministry of Climate Change, Government of Pakistan, 2022). In the interim the international gridded products—EDGAR (Crippa et al., 2025, 2020) and Climate TRACE (Climate TRACE, 2026)—supply the numbers that inform policy. They are global models applying globally consistent assumptions to nationally reported activity, which is their strength for cross-country comparison and their limitation for any one country: where a national compiler can consult a sanitation survey or a district population panel, a global product must choose a default.

The waste sector is where that difference bites hardest, for two reasons. Its activity data are the least standardised internationally, so the defaults do the most work. And it is dominated by methane, whose short atmospheric lifetime makes it the gas where near-term mitigation is most effective and therefore where the estimate most directly changes what a government should do.

### *1.3. Contributions*

1. **A current national waste-sector inventory for Pakistan** covering solid-waste disposal, open burning and domestic wastewater, from a district population panel and national sanitation statistics rather than from a global model's Pakistani slice.
2. **One waste stream, mass-balanced across its fates**: collected and uncollected shares sum to generation exactly, two fates carry emissions pathways (4A, 4C2) and the unburned remainder ships as an explicit unresolved row rather than a silent omission. The closed balance removes a class of inconsistency that independent estimation cannot detect (Section 4).
3. **Both readings of the sewer methane correction factor**, with the evidence for each stated and neither resolved by fiat, together with the measurement that would settle it (Section 9).
4. **A spin-up sensitivity for the decay model**, quantifying how much of today's landfill methane comes from waste deposited before the activity panel begins—a question most first-order-decay implementations answer implicitly and do not report.
5. **An air-pollutant layer on the same activity basis** as the greenhouse gases, so that open burning's particulate can be compared with other Pakistani sources without the usual mismatched-activity problem.

Method that this sector shares with the other nine of the national inventory—the module contract, the uncertainty treatment, the notation keys, the accounting closures—is documented in the national manuscript (Ahmad, 2026) and not repeated here.

## 2. Related Work

### *2.1. Global products and what they can and cannot resolve*

EDGAR (Crippa et al., 2025, 2020) reports a Pakistani waste total covering disposal, burning and wastewater together, and is the closest external comparator to the scope reported here. Climate TRACE (Climate TRACE, 2026) takes a different approach, inferring emissions from asset-level and remote-sensing evidence. Both are global products: they apply a consistent methodology across every country, which is what makes them comparable and also what limits them nationally. A global waste model must assume a collection rate and a sewer condition for Pakistan; a national compiler can consult the sanitation statistics and the district population panel that describe it.

The disagreement between products is not a defect of any of them, but it is large enough to matter. This inventory sits at 141 % of EDGAR's year-matched Pakistani waste figure, and Section 6 argues that a single assumption—the sewer methane correction factor—accounts for most of the difference.

### 2.2. *Waste generation and composition in South Asia*

Per-capita waste generation and its composition are the two activity inputs the sector cannot avoid, and both are surveyed rather than metered. The global compilation by Kaza et al. (2018) is the standard reference; its Pakistan generation row re-bins the one primary national survey, the Pak-EPA eight-city study (Pakistan Environmental Protection Agency and Environmental Protection and Management Consultants, 1996), whose rates are used here directly, and its collection coverage supplies the uncollected fractions. Composition matters more than it first appears: the fossil carbon fraction of municipal waste—plastics, textiles, rubber—determines how much of the $CO_2$ from open burning is reportable, because the biogenic remainder is a memo item under IPCC's convention rather than a national-total emission.

### 2.3. *First Order Decay and its implementation choices*

Landfill methane is not a function of this year's waste but of every year's waste still decaying, which is why IPCC's Volume 5 method is a first-order decay model over a deposit history rather than an emission factor applied to an annual tonnage. Three implementation choices then do most of the work and are reported inconsistently in the literature: the decay constant, the methane correction factor describing site management, and how far back the deposit series is extended before the model is considered spun up.

The third is the one this paper treats explicitly. A model started at the first year of an activity panel implicitly asserts that nothing was landfilled before it, which is never true and always biases the estimate low. The size of that bias is rarely reported. Section 7 reports it here.

### 2.4. *The wastewater methane correction factor*

For domestic wastewater the corresponding controversy is the methane correction factor, which translates a sanitation pathway into the fraction of its organic load that degrades anaerobically. IPCC's default table distinguishes stagnant sewers from flowing ones by an order of magnitude in that fraction, and the distinction was drawn for systems that are clearly one or the other. Open urban drains that are stagnant in the dry season and flowing after rain, and which discharge to irrigation channels, are not obviously either.

Pakistani sanitation coverage itself is well characterised: the WHO/UNICEF Joint Monitoring Programme (WHO/UNICEF Joint Monitoring Programme, 2023) reports the population shares using sewers, septic systems, latrines and open defecation, and those shares are what allocate the national organic load across pathways here. What the JMP does not report, and no Pakistani source does, is the hydraulic condition of the sewered fraction—which is the quantity the correction factor is asking about.

## 3. Study Area and Data

### 3.1. *Population, which is the activity datum*

Every category in this sector is driven by population. Waste generation is a per-capita rate applied to urban and rural residents; the organic load entering the wastewater system is a per-capita BOD applied to the same population split by sanitation pathway. There is no tonnage measurement and no metered flow anywhere in the chain, so the population panel is the activity data in a way that is unusual for an emissions inventory.

The population panel covers 134 district units in the four provinces and Islamabad Capital Territory, using the census-anchored interpolation and explicitly labelled projections. Gilgit-Baltistan and Azad Kashmir are not individually represented in that panel. The agricultural allocation uses 116 four-province districts plus its grouped Other residual; the residential summary also groups Other rather than placing Islamabad individually. The reference population resolves to 97 million urban and 150 million rural residents, 247 million in total. The urban/rural split is load-bearing twice over: urban residents generate more waste per head and are far more likely to have their waste collected, and the two effects run in opposite directions for emissions, since collected waste goes to a decaying landfill while uncollected waste burns.

Building the panel at district level rather than nationally is what makes the provincial allocation in Section 7 a real geography rather than a national figure split by a population share, and it is the same panel that drives the sector's air-pollutant layer.

### 3.2. *Waste generation and collection*

Per-capita generation is 0.453 kg per person per day urban and 0.283 kg rural, from the Pak-EPA eight-city survey (Pakistan Environmental Protection Agency and Environmental Protection and Management Consultants, 1996), the rates the Kaza et al. (2018) Pakistan row re-bins. Across the panel that is 31.56 Mt of municipal solid waste a year.

Collection is expressed as its complement, the uncollected fraction: 30 % urban and 70 % rural, which resolves to 50.3 % of national generation collected. Of the uncollected remainder, 60 % is openly burned; the rest is dumped, tipped into watercourses or otherwise disposed of without combustion and emits nothing this category reports.

These three parameters—two generation rates, two uncollected fractions and one burned fraction—are the softest inputs in the sector, and Section 10 says so. They are also the only place where a judgement moves emissions *between* categories rather than changing the total, which is a property of the partition rather than of the parameters.

### 3.3. *Waste composition*

Open burning needs the carbon content of what burns. Dry matter is 70 % of wet weight; total carbon is 0.2195 t per wet tonne and the fossil fraction of it is 0.0757 t per wet tonne, so roughly a third of the carbon in Pakistani municipal waste is fossil in origin. That third is what produces reportable $CO_2$; the biogenic remainder is a memo item, following IPCC Volume 5 Chapter 5 §5.1.

### 3.4. Sanitation

The wastewater category needs the population split across sanitation pathways, because the methane correction factor is a property of the pathway rather than of the person. Shares come from the WHO/UNICEF Joint Monitoring Programme (WHO/UNICEF Joint Monitoring Programme, 2023), each year taking its own: sewered, septic, latrine and open defecation, each carrying its own correction factor (0.5, 0.5, 0.1 and 0 respectively).

Organic load is 40 g BOD per person per day, the regional default covering Pakistan, with an industrial-and-commercial correction of 1.25 for load discharged to the domestic system from small commercial premises. Nitrous oxide is computed from protein consumption rather than from BOD, which is why it is invariant under both of the sensitivities reported in Section 7.

### 3.5. What this inventory has not obtained, and what it would change

Nothing compiled here reports landfill tonnages at facility level, methane recovery at any site, industrial solid-waste quantities, or industrial wastewater organic loading. Three of those four have leads that have not been followed: the Urban Unit has mapped solid-waste services and facilities across Punjab's tehsils, PBS's district-level living-standards survey records household disposal route—municipal collection, private, public bins, roadside, waterway, open space—and PCRWR has published organic loading for named Pakistani drains. None would give complete national coverage, and each would replace a national assumption with district or facility evidence for the population it covers. The first two would replace modelled disposal with measured disposal; the second two are whole categories this inventory reports as not estimated. Section 10 takes each in turn.

## 4. Methods

### 4.1. One generation figure, mass-balanced across fates

The sector's structure follows from a single decision: municipal solid waste is one national quantity that must be divided, not three quantities to be estimated separately.

Annual generation is

$$G \;=\; 365 \sum_{d} \left( P_d^{u} g_u + P_d^{r} g_r \right), \tag{1}$$

over districts $d$, with $g_u$ and $g_r$ the urban and rural per-capita rates. The collected fraction, which is what reaches a disposal site,

$$D \;=\; 365 \sum_{d} \left( P_d^{u} g_u (1 - u_u) + P_d^{r} g_r (1 - u_r) \right), \tag{2}$$

uses the uncollected fractions $u_u$ and $u_r$, and the uncollected remainder is $U = G - D$ by construction rather than by a second estimate. Of that remainder, the burned quantity is $B = f_b U$.

The identity $D + U = G$ is asserted in code with exact equality, and a test fails if it ever ceases to hold. (Tier status is stated per component rather than per sector, in Appendix E: the categories share one country-specific activity basis but differ in how much of each calculation is a default.) This is the property that separate estimation cannot provide. If landfill tonnage comes from one source and open-burning tonnage from another, their sum is unconstrained and can exceed or fall short of national generation without anything in the calculation noticing; here, an argument about the collection rate moves emissions between 4A and 4C2 and changes the total only through the different methane yields of the two fates.

For FY2023–24 this gives 31.56 Mt generated, 15.88 Mt collected to disposal, 15.67 Mt uncollected and 9.40 Mt openly burned. The partition is published as an executable fate ledger (`outputs/waste_ledger_2024.csv`, the same claimant structure as the transport fuel ledger), whose writer raises if either tier fails to close and whose final row is the fate this inventory does not model: 6.27 Mt of uncollected waste that does not burn—dumped on land or tipped into watercourses, carried explicitly rather than silently.

### 4.2. Solid-waste disposal: First Order Decay (4A)

Methane from a disposal site in year $t$ depends on every year's deposit still decaying, so IPCC Volume 5 Chapter 3's first-order decay model is run over a deposit history rather than applied to a single year's tonnage. Decomposable degradable organic carbon accumulates and decays as

$$\mathrm{DDOCm}_a(t) = W(t)\,\mathrm{DOC} \cdot \mathrm{DOC_f} \cdot \mathrm{MCF} + \mathrm{DDOCm}_a(t-1)\, e^{-k}, \tag{3}$$

$$\mathrm{CH_4}(t) = \left[ \mathrm{DDOCm}_a(t-1) \left( 1 - e^{-k} \right) \right] F \cdot \tfrac{16}{12}, \tag{4}$$

with $k = 0.065$, $\mathrm{DOC_f} = 0.5$, $\mathrm{MCF} = 0.6$ and $F = 0.5$.

Two parameters are deliberately zero. No methane recovery is credited, because none is documented at national scale and crediting an undocumented recovery would understate the category. The oxidation factor is zero for the same class of reason: IPCC's non-zero oxidation applies to managed sites with a cover layer, and Pakistani disposal is predominantly unmanaged, where the thin intermittent cover that drives oxidation is absent.

#### 4.2.1. Spin-up, and why it is reported rather than assumed

The activity panel begins in 2000, but waste deposited before then is still emitting. Starting the model at the panel's first year with an empty site would assert that Pakistan landfilled nothing before 2000, which is false and biases the estimate low.

The deposit series is therefore back-projected to 1960 at 40 % of the panel's first-year tonnage, and the model is run from there. Because that back-projection is an assumption rather than data, the sector reports what it is worth: without spin-up the category would be 8.07 Mt instead of 9.17 Mt, so pre-2000 deposits contribute 12 % of today's landfill methane. A reader who prefers to start at the panel can read the alternative directly from the output file rather than recomputing it.

#### 4.2.2. *The monthly state engine, and why it is one recursion*

The national release resolves the disposal stream into fiscal months, and the engine behind those months is this model run at a monthly step rather than a reshaped annual answer: one continuous single-pool recursion from the 1960 spin-up, $S_m = S_{m-1}\, e^{-k\, d_m/365.25} + W_m$, where $d_m$ is the month's exact calendar length (leap Februaries included) and $W_m$ deposits the month's own calendar year's tonnage, days-weighted within the year. Running one recursion rather than a per-year replay matters for three named reasons: state flows across fiscal boundaries (June's stock is July's opening balance, so no seam appears at the year change); each month deposits its *own* year's waste (a per-year replay had injected the panel's maximum year into every year's stock); and the reaction delay between deposit and the onset of decay is a declared parameter (`swds_delay_months`, the IPCC six-month convention) applied as a shape refinement only. The single pool is kept deliberately — the annual model carries one decay constant, and a cohort structure would claim a waste-composition split no Pakistani record supports. The annual control is preserved exactly by construction: each fiscal year's twelve shares are normalised against its own annual figure, so the monthly engine decides only *when* within the year, never how much, and the within-year spread it computes is printed into the month rows' basis rather than asserted in prose.

### 4.3. *Open burning (4C2)*

IPCC Equation 5.1 applied to the burned quantity $B$, taking that chapter's own allowance to combine the dry-matter, carbon-content and fossil-fraction terms into one coefficient rather than carrying them separately:

$$\mathrm{CO_2} = B \cdot \mathrm{CF_{fossil}} \cdot \mathrm{OF} \cdot \tfrac{44}{12}, \tag{5}$$

where $\mathrm{CF_{fossil}}$ is the fossil carbon per *wet* tonne given in Section 3 and the oxidation factor is OF = 0.71. Dry matter does not appear here because it is already inside that coefficient; it enters this category only through the $N_2O$ factor, which IPCC states per dry tonne. The burned quantity $B$ is itself IPCC Equation 5.7.

Only the fossil carbon fraction produces reportable $CO_2$. The biogenic fraction is reported as a memo item at 3520 Gg $CO_2$, outside the national total, because that carbon was fixed from the atmosphere within the year. Methane and nitrous oxide from the same burning *are* counted regardless of carbon origin: incomplete combustion is a net source however the carbon entered the waste stream.

### 4.4. *Domestic wastewater (4D1)*

Tier 1 over the sanitation-resolved population. Total organically degradable material is

$$\mathrm{TOW} = P \cdot \mathrm{BOD} \cdot 365 \cdot I, \tag{6}$$

with $I = 1.25$ the industrial-and-commercial correction, and methane is

$$\mathrm{CH_4} = \sum_j \mathrm{TOW}_j \cdot B_0 \cdot \mathrm{MCF}_j, \tag{7}$$

summed over sanitation pathways $j$ with $B_0 = 0.6$ kg $CH_4$ per kilogram BOD. The pathway correction factors are the subject of Section 9.

Nitrous oxide is computed from protein consumption rather than from BOD, by the method of the 2019 Refinement (Volume 5 Chapter 6, Equations 6.7 to 6.10a) (IPCC, 2019). Protein supply is reduced to protein consumed (a factor of 0.96) and its nitrogen (0.16) is allocated to the same sanitation pathways the methane uses, with factors of 1.1 for household products, 1.1 for non-consumed protein (the chapter's updated Table 6.11, taken over its text, which gives no increment where food waste goes with solid waste) and, for industrial and commercial co-discharge, 1.25 on the sewered population only. Treatment removes part of each pathway's nitrogen before discharge—0.15 in septic tanks, 0.12 in latrines and 0.40 in the treated share of sewered flow, which is declared at 1 % and not measured—and the nitrogen entering a treatment plant emits there at 0.016. What is left is summed over every pathway and emits at the effluent factor of 0.005; the whole population stays in the estimate. The result is 9.71 Gg of $N_2O$. The single equation of the 2006 Guidelines, which this inventory used before and which treats one nitrogen total on protein supply as effluent, is retained as the alternative and gives 11.45 Gg. Neither method reads BOD or the sewer correction factor. This is why nitrous oxide does not move under either of the sensitivities reported below, and why a sensitivity row that reported methane alone would be misread as a category total—a distinction this sector's output file previously got wrong and now states explicitly.

## 5. Emission Factors and Parameters

Every factor and parameter used here is an IPCC 2006 default, a 2019 Refinement default (the wastewater nitrous oxide chain) or a documented regional value, with its source recorded in the versioned parameter table. Three deserve discussion because the choice was not automatic.

### 5.1. *Two oxidation factors that must not be reconciled*

This sector carries an oxidation factor of 0.71 for open burning and 0 for landfill, while the fossil-fuel sectors of the same inventory carry none at all. That looks like an inconsistency and is not.

The 2006 Guidelines fold complete oxidation into the fuel combustion emission factors themselves—Table 1.4 carries an explicit oxidation column of 1.0 for every fuel—so applying the 1996-vintage 0.98 on top of a 2006 factor double-discounts. Open burning is different in kind: it genuinely is incomplete, and 0.71 is the 2006 value that says so. Landfill oxidation is different again, describing methane oxidised while passing through a cover layer, which unmanaged Pakistani sites largely do not have.

Three different numbers describing three different physical situations. Tests pin all three, so that a future reader who notices the apparent inconsistency and "corrects" it fails the build rather than silently changing the sector.

| Parameter | Applies to | Value |
|---|---|---|
| Generation rate, urban | 4A, 4C2 | 0.453 kg/cap/day |
| Generation rate, rural | 4A, 4C2 | 0.283 kg/cap/day |
| Uncollected fraction, urban | 4A, 4C2 | 0.30 |
| Uncollected fraction, rural | 4A, 4C2 | 0.70 |
| Burned fraction of uncollected | 4C2 | 0.60 |
| Decay constant $k$ | 4A | 0.065 $yr^{-1}$ |
| $DOC_f$ | 4A | 0.50 |
| MCF (site management) | 4A | 0.60 |
| $F$ ($CH_4$ in landfill gas) | 4A | 0.50 |
| Oxidation factor | 4A | 0 |
| Methane recovery | 4A | 0 |
| Back-projection to 1960 | 4A | 40% of first panel year |
| Dry matter | 4C2 | 0.70 |
| Total carbon | 4C2 | 0.2195 t C/t wet |
| Fossil carbon | 4C2 | 0.0757 t C/t wet |
| Oxidation factor | 4C2 | 0.71 |
| BOD | 4D1 | 40 g/cap/day |
| $B_0$ | 4D1 | 0.6 kg $CH_4$/kg BOD |
| MCF, sewered / septic | 4D1 | 0.50 |
| MCF, latrine / open | 4D1 | 0.10 / 0 |
| Industrial-commercial correction | 4D1 | 1.25 |

Table 1: Parameters, with the IPCC volume and table recorded against each in `data/waste_params.csv`. The two zeros in the 4A block are deliberate and argued in Section 4, not defaults left unset. Full row-level provenance—value, uncertainty, source and basis for every parameter and factor—is the generated table in Appendix F.

### 5.2. *Fossil carbon, and why only a third of the burning $CO_2$ is reported*

Total carbon in Pakistani municipal waste is 0.2195 t per wet tonne, of which 0.0757 t—roughly a third—is fossil. Only that third produces reportable $CO_2$; the rest was fixed from the atmosphere by plants within the year and is reported as a memo item at 3520 Gg.

This convention is easy to apply inconsistently, because the same waste produces $CH_4$ and $N_2O$ that *are* counted in full regardless of carbon origin. The distinction is not about the waste but about the atmosphere: biogenic $CO_2$ returns carbon that was recently removed, while methane from the same carbon is a different molecule with a different radiative effect and no such offset.

### 5.3. *What the parameters are worth*

Three of the parameters above dominate the sector's uncertainty, and each is examined as a published sensitivity rather than left implicit: the sewer MCF (Section 9), the BOD rate, and the spin-up back-projection. The generation and collection rates are not among them—not because they are well known, but because the partition means they move emissions between categories more than they move the total.

## 6. Validation

There is nothing to validate this sector against directly. No Pakistani facility reports landfill gas, no drain is monitored for dissolved methane, and no survey measures how much uncollected waste burns. What exists is other people's models, and comparing a model against a model establishes consistency rather than accuracy. This section says what that comparison does and does not show.

### 6.1. *Against EDGAR*

EDGAR reports Pakistani waste—disposal, burning and wastewater together—at 26.57 Mt $CO_2e$ for 2023, year-matched to this inventory's reference year (Crippa et al., 2025). This inventory reports 37.45 Mt, or 141 % of it. (EDGAR's 2024 value, 27.21 Mt, is carried in the benchmark file for release-vintage context and is not the comparison basis.)

That row has changed sides during this work, which is what makes it informative. Before domestic wastewater was built, this sector covered disposal and burning only and read 49 % of EDGAR; the gap was then a measure of what the missing category would add. With 4D1 in, the sector overshoots. The same comparison has therefore diagnosed an omission and then a possible overestimate, which is more than a single ratio usually earns.

The likeliest single cause is the sewer methane correction factor. At IPCC's stagnant-sewer 0.5 the wastewater category is 24.46 Mt and the sector overshoots; at the river-discharge 0.1 it is 15.39 Mt, the sector is 28.38 Mt, and the comparison lands at 1.07 times EDGAR. No other single parameter in the sector can move it that far.

It would be easy to treat that as confirmation and adopt 0.1. We do not, and Section 9 argues why: EDGAR is itself a model making its own assumption about the same unmeasured quantity, so agreement between the two would demonstrate that two models had made the same choice rather than that the choice is right.

### 6.2. *Against Climate TRACE*

Climate TRACE's Pakistani waste subsectors are built from a different evidence base again, including facility detection. Its solid-waste figures are broadly consistent with the disposal category reported here; its coverage of open burning and domestic wastewater is not constructed on a comparable scope, so a total-to-total comparison would mislead rather than inform and is not reported as one.

### 6.3. *Against household survey evidence*

The 2024–25 Household Integrated Economic Survey asks every sampled household how its waste is collected or disposed of (Pakistan Bureau of Statistics, 2025), and its design-weighted answers are the first Pakistani survey evidence bearing on the partition's load-bearing parameters. They disagree with the What a Waste collection coverage the partition is built on, in one direction: 42.0 % of urban households report no collection of any kind against the 30 % assumed, and 93.4 % of rural households against 70 %. The route detail supports the open-dumping reading of the uncollected stream—88.5 % of rural households report disposal to open space, and under 1 % to waterways.

Two caveats keep this a comparison rather than a correction. Household route shares are not waste-mass shares: the survey weights households, not tonnes, and converting one to the other needs stratum generation rates it does not carry. And counting public-bin use as collected overstates collection where bins are cleared infrequently, which the same module records. The configuration is therefore priced as a named structural scenario in the supplement rather than adopted; because the partition conserves generated waste, moving the uncollected fractions to the survey values moves emissions from slow landfill decay to immediate burning without changing the waste the country produces.

### *6.4. Internal checks that do have teeth*

Three checks are enforced on every run and, unlike the external comparisons, can fail.

**The mass balance closes.** Disposed plus uncollected equals generated, exactly. This cannot detect a wrong collection rate, but it can and does detect the class of error where a change to one category silently fails to appear in the other.

**The burned quantity is exactly the burned fraction of the uncollected remainder.** Asserted with equality against an independently recomputed value, so a future refactor that accidentally applies the burned fraction to total generation rather than to the uncollected part fails immediately. That specific error would inflate open burning by a factor of two and is entirely plausible.

**Provincial allocation reconciles exactly.** The four provinces and the territories sum to the national figure for each category, not merely for the sector.

### *6.5. What would constitute real validation*

Three measurements, in order of what they would settle. Dissolved-methane and flux sampling on urban drains across a seasonal cycle would resolve the correction factor that dominates the sector. Landfill gas monitoring at two or three large sites would test the first-order-decay parameters directly. A waste-characterisation survey at the same sites would replace the regional composition figures with Pakistani ones and firm up the fossil carbon fraction that governs reportable $CO_2$ from burning.

None is expensive relative to what the sector contributes to national uncertainty. That argument is made quantitatively in Section 9.

## 7. Results

### *7.1. The sector*

Pakistan's waste sector emitted 37.45 Mt $CO_2$e in FY2023–24 on AR5 metrics, or 36.37 Mt on AR6. Domestic wastewater is 65.3 % of it.

Methane is 87 % of the sector, which is why the AR5-to-AR6 difference of 1.08 Mt is almost entirely a methane effect and why this sector is unusually sensitive to the metric chosen.

| Category | Gas | Gg | Mt $CO_2$e | ±% |
|---|---|---|---|---|
| Open burning (4C2) | $CO_2$ (fossil) | 1853.21 | 1.853 | 84.3 |
| | $CH_4$ | 61.12 | 1.712 | 92.9 |
| | $N_2O$ | 0.99 | 0.262 | 124.2 |
| Solid-waste disposal (4A) | $CH_4$ | 327.45 | 9.169 | 79.8 |
| Domestic wastewater (4D1) | $CH_4$ | 781.63 | 21.886 | 73.7 |
| | $N_2O$ | 9.71 | 2.572 | 268.0 |
| **Total** | | | **37.453** | **51.1** |

Table 2: Waste-sector emissions, FY2023–24, AR5 GWP-100. Biogenic $CO_2$ from the biomass content of burned waste, 3520 Gg, is a memo item outside the total per Vol 5 Ch 5 §5.1. The sector band is the Approach-1 quadrature over these six components.

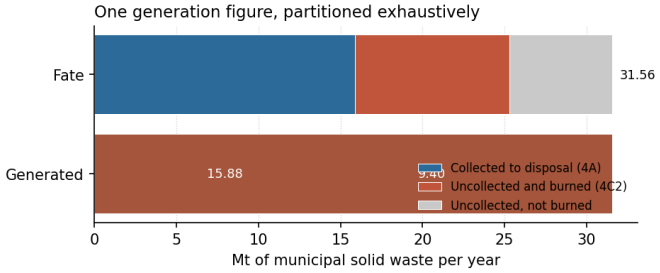



Figure 1: The partition. 31.56 Mt of municipal solid waste is generated; 15.88 Mt is collected and enters the decay model as 4A; 15.67 Mt is uncollected, of which 9.40 Mt burns as 4C2 and the remainder is dumped without combustion. The two lower bars sum to the upper one exactly, which is the property the code asserts rather than checks.

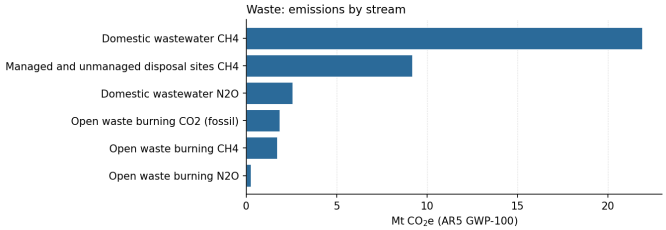



Figure 2: Emissions by stream. Domestic wastewater methane alone, at 21.89 Mt $CO_2$e, is larger than the other five streams combined.

### *7.2. Where the uncertainty is*

The factor and activity uncertainties differ substantially by pathway. The current common ensemble reruns the native waste calculation for each realisation and preserves both the waste-fate identity and the asymmetric effluent-$N_2O$ factor distribution. Its reference-year conditional interval is 25.4 Mt to 61.5 Mt $CO_2$e on AR5. This is a parameter interval within the selected management interpretation and source coverage; structural cases remain separate. The standalone sector ensemble and Approach 1 screen retain their own stated settings and are not substituted for the common release interval (Supplementary Note S1).

### *7.3. Sensitivities, published rather than chosen*

The spin-up row answers a question most first-order-decay implementations answer implicitly: pre-2000 deposits contribute 12 % of today's landfill methane. An implementation

| Assumption | Category (Mt) | Sector (Mt) |
|---|---|---|
| Sewer MCF 0.5 (reported) | 24.46 | 37.45 |
| Sewer MCF 0.1 (river reading) | 15.39 | 28.38 |
| BOD at India's 34 g/cap/day | 21.17 | 34.17 |
| Disposal with spin-up (reported) | 9.17 | 37.45 |
| Disposal without spin-up | 8.07 | 36.35 |

Table 3: The three assumptions this sector cannot settle, each reported as an alternative rather than absorbed into the central estimate. Nitrous oxide is invariant under both wastewater cases because it derives from protein rather than from BOD or sewer condition, so all rows are category totals including it—a distinction this sector's own output file previously got wrong.

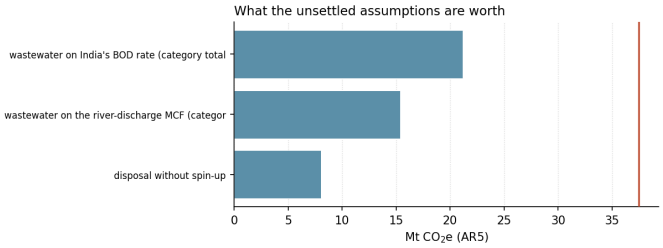



Figure 3: The same result as a range around the reported figure. A reader who has to reconstruct a range from prose will not reconstruct it.

that starts at its activity panel is not making no assumption; it is asserting that the country landfilled nothing before its data begin.

Named structural cases retain their own sewer and management assumptions. They are not combined into a probability mixture. Conditional parameter intervals describe a specified case; differences between cases remain a separate sensitivity, including the alternative partition based on the HIES household route shares in Section 6.

### 7.4. Provincial allocation

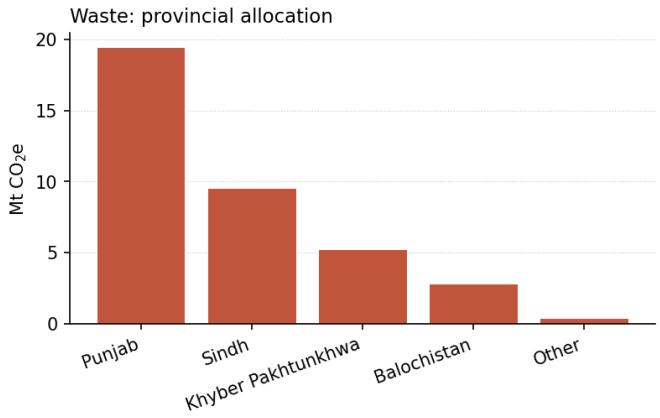



Figure 4: Punjab 19.49 Mt, Sindh 9.55 Mt, Khyber Pakhtunkhwa 5.21 Mt, Balochistan 2.80 Mt and 0.40 Mt in the territories. Allocated on the same district population panel that drives generation. This is an *origin* allocation: population locates where waste and wastewater are produced, and the split reconciles exactly against each category's national figure—but it carries the further assumption that release is co-located with origin, which is the stronger claim.

The provincial pattern follows population closely, which is expected for a sector whose activity datum *is* population, and is the reason this sector's geography is less informative than, say, the power sector's. What the allocation locates is the *origin* of the waste and wastewater—where the generating people live—not the release: disposal methane is emitted at landfills and dumps, wastewater methane along drains, at treatment works and at discharge points, and burning at burn sites. What the inventory holds toward that release geography is a prospective seed only: 56 OpenStreetMap `landuse=landfill` polygons (OpenStreetMap contributors, 2026) (ODbL 1.0 derivative database, `data/gis/landfills/`), 13 of them named, quality-flagged, and touched by no estimated stream—far too sparse for an SWDS allocation, and registered so the release can cite what it holds rather than claiming it holds nothing. The three objects are published SEPARATELY, as the review asked: origin (`waste_district_2024.csv`), fate (the ledger), and release—the polygons republished as candidates (`outputs/waste_release_candidates_2024.csv`) with an assignment product (`outputs/waste_release_assignment_2024.csv`) whose defining column is EXPLICIT UNASSIGNED MASS: a district's disposal emissions are assigned only to its own mapped candidates (equal split, capacity unknown and declared), nothing crosses a district boundary, and 75 % of the disposal category stays unassigned across 113 districts—the honest measure of how far a release map is from existing. At provincial resolution the origin/release distinction is modest, because waste rarely crosses a provincial boundary between generation and fate; at any finer resolution the population surface would place landfill methane in residential neighbourhoods, which is why the sector's ALLOCATION remains origin-based and the limitations section carries the gap.

### 7.5. Contribution to the national inventory

The category composition separates wastewater methane and $N_2O$, disposal-history methane and burning releases. Their uncertainty and user relevance depend on the species: a small contribution to the GHG aggregate does not imply a small primary PM or other pollutant release. The current native ensemble and pollutant-specific driver diagnostics are used to identify parameter sensitivity, without treating the old Approach 1 category screen as an additive variance decomposition of the nonlinear national ensemble.

## 8. Air-Pollutant Co-Emissions

Open municipal-waste burning uses the same wet/dry burned quantity and fate account as the greenhouse-gas calculation. The FY2023–24 reference burned mass is 9.40 Mt; pollutant-specific garbage-burning factors follow Andreae (2019). Masses are reported in native units. Burning locations are not observed nationally, so the population allocation describes waste origin rather than confirmed release sites.

### 8.1. Interpretation of primary particulate releases

The reference estimate is 63.85 kt of primary $PM_{2.5}$. Its relative contribution depends on the compared pollutant, source

| Pollutant | kt/yr |
|---|---|
| CO | 434.45 |
| NMVOC | 151.40 |
| $PM_{2.5}$ | 63.85 |
| HCl | 18.43 |
| $NO_x$ (as $NO_2$) | 15.14 |
| Black carbon | 9.22 |
| $NH_3$ | 5.92 |
| $SO_2$ | 3.29 |

Table 4: Air pollutants from open waste burning, FY2023–24. Mass only; these never enter a greenhouse-gas total. Computed from the same burned tonnage as the 4C2 emissions above, so the two describe the same fire. $NO_x$ is reported as $NO_2$-equivalent (Andreae publishes the garbage-burning factor as NO; the conversion is documented on the factor row), the inventory-wide basis.

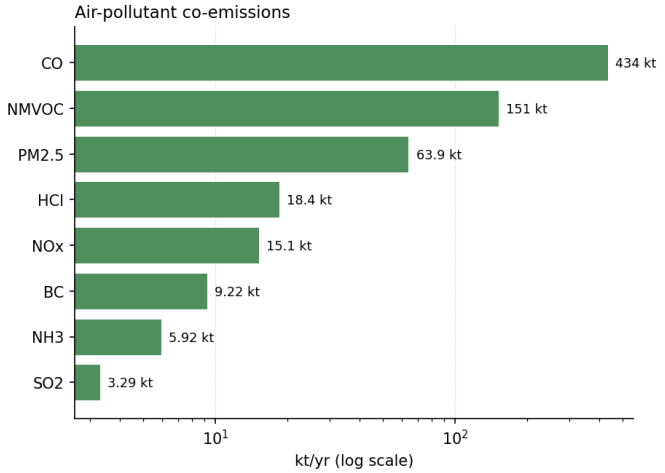



Figure 5: The same figures on a log axis, which the range demands: carbon monoxide and sulphur dioxide differ by more than two orders of magnitude, and a linear axis renders everything below $PM_{2.5}$ as a line of zero width.

coverage and household or industrial configuration. No inverse relationship between GHG and pollutant rankings is assumed. Emitted mass does not establish exposure: disposal and burning locations, release conditions, atmospheric transport and chemistry remain outside this calculation.

The common activity account prevents disposal and burning from claiming the same generated waste twice. It does not establish where the uncollected material burns or validate the pollutant factors. The unburned uncollected fraction remains an unresolved fate, rather than an invented combustion destination.

### *8.2. Black carbon and hydrogen chloride*

Two entries deserve comment because they are specific to waste rather than generic to combustion.

Black carbon at 9.22 kt is a strong short-lived warming agent that this inventory reports as mass and does not convert to $CO_2e$, following the convention that its metrics remain contested. It is reported because omitting it would understate the climate relevance of open burning even while the greenhouse-gas total for 4C2 is small.

Hydrogen chloride at 18.43 kt appears in no other sector of this inventory and is a direct consequence of burning chlorinated plastics in the open. It does not quantify dioxin or furan releases, which require separate source-specific methods.

### *8.3. Limitations of this layer*

The conditional air ensemble reuses the same bounded waste-fate realisations as the GHG calculation, then applies pollutant-factor uncertainty. Burning-fraction alternatives remain separate structural cases; their endpoints are not confidence limits. The composition is regional rather than Pakistani, which matters more for pollutants than for greenhouse gases because plastics and textiles drive the chlorine and much of the particulate. And the factors describe open burning generically rather than the specific conditions of Pakistani street-side burning—smouldering, damp, and often mixed with organic waste—whose consequences for incomplete-combustion products require local evidence.

## 9. Discussion

### *9.1. The methane correction factor, and why it is not resolved here*

The sewer methane correction factor is the largest single assumption in this sector and the one the evidence does not settle. It deserves the space because how it is handled is a methodological position, not a parameter choice.

IPCC gives 0.5 for stagnant open sewers and 0.1 for wastewater discharged to a flowing, well-aerated body. The distinction was drawn for systems that are clearly one or the other. Pakistani urban drainage is not: the drains are open, warm, slow-moving and heavily organically loaded, which is the description 0.5 exists for, while a substantial fraction ultimately discharges to rivers and irrigation channels, which is the description 0.1 exists for. Both descriptions are accurate about different parts of the same system, and no Pakistani source reports the hydraulic condition of the sewered fraction, which is the quantity the factor is actually asking about.

The consequence is a factor of 1.59 on the sector's largest category: 24.46 Mt against 15.39 Mt, or 37.45 Mt against 28.38 Mt for the sector.

#### *9.1.1. Why external agreement is not a tiebreak*

At 0.1 this sector lands close to EDGAR, and it would be easy to present that as corroboration. It is not, and treating it as one would be a methodological error worth naming because it is a tempting one.

EDGAR is a model. Its Pakistani wastewater figure rests on its own assumption about the same unmeasured quantity—the anaerobic fraction of Pakistani sewage—made from the same absence of Pakistani measurement. If this inventory adopted 0.1 because it produces agreement, the two would agree because both had chosen the same default, and the agreement would then be cited as though it were evidence about Pakistani drains. That is how a shared assumption becomes an apparent consensus.

We therefore report 0.5 as central, because the physical description of the drains is the stronger evidence and is specific

to Pakistan, and publish 0.1 beside it so a user who weighs the comparator differently can take it. What we do not do is let the comparison choose.

#### *9.1.2. What would settle it*

Dissolved-methane concentration and flux measurement on urban drains in two or three Pakistani cities, sampled across a seasonal cycle to capture the dry-season stagnation and the monsoon flushing that the two IPCC categories describe separately.

Domestic wastewater is consequential because both the receiving environment and the effluent emission factor remain uncertain. Seasonal methane and nitrogen-pathway measurements could distinguish these interpretations. An old Approach 1 variance ranking is not used as a current nonlinear variance budget or a claim about the cost-effectiveness of new measurements.

### *9.2. What the closed mass balance buys, and what it does not*

The balance guarantees that the collected and uncollected streams together account for exactly the waste the country generates, with the burned quantity a declared fraction of the uncollected stream and the unburned 6.27 Mt carried as an explicit unresolved fate rather than assigned to either category. That removes a specific class of error—the two categories drifting into mutual inconsistency as one is updated—and makes the collection rate an explicit, single, arguable number rather than an implicit consequence of two independent estimates.

It is worth being precise about what it does not buy, because a closed balance invites overconfidence. It says nothing about whether the split point is right. The collection rate of 50.3 % carries real uncertainty, and moving it moves waste between two fates with quite different methane yields per tonne: a tonne landfilled decays slowly and anaerobically over decades, while a tonne burned emits immediately and mostly as $CO_2$. Nor does it validate the generation figure itself; a partition of a wrong total still closes.

What it changes is the shape of the argument. A reader who disagrees with the collection rate can see exactly which two categories move and in which directions, rather than having to work out whether two independently sourced numbers are mutually consistent.

### *9.3. Implications*

The principal implications follow the material pathways.

**Wastewater fate is a consequential GHG uncertainty.** Sewer condition, receiving environment and effluent pathways require distinct evidence. Burning composition and location constrain different outputs, especially primary particulates and other air pollutants, and are not dismissed on the basis of the wastewater share of $CO_2$ equivalent.

**Open burning requires pollutant-specific interpretation.** The shared activity account supports primary particulate and other combustion pollutant estimates alongside GHGs. These mass estimates identify the importance of the assumed burning fate within the covered source panel. They do not establish exposure or the benefits of an abatement programme without release-location, transport and population evidence.

**The collection rate is a policy variable, not just a parameter.** Because the partition makes it the single number governing what fraction of waste burns rather than decays, improving collection is visible in this inventory as a movement between categories. That is the right structure for evaluating a municipal collection programme, and it is not available in an inventory whose two categories are estimated independently.

## 10. Limitations

**The sewer methane correction factor is unresolved, and it is the sector.** A factor of 1.59 on two-thirds of the emissions, argued in Section 9. Both readings are published; neither is measured.

**Industrial solid waste and industrial wastewater are not estimated at all.** No national quantity for either has been compiled here. The categories reported here are municipal and domestic only, and both gaps carry the notation key NE with that justification, in the complete sector-4 matrix of Appendix G—which also declares biological treatment (4B) and incineration (4C1) as NE rather than leaving them silently absent. Industrial wastewater is the more consequential omission: given that domestic wastewater alone is 5.5 % of the national inventory, an industrial stream from a textile and tanning sector of Pakistan's size is unlikely to be negligible.

**Generation rates are single national values.** The urban and rural per-capita rates do not vary by province or city size, which is a real simplification for a country whose largest city and smallest districts differ substantially in consumption and in what they discard. The rates are regional literature values rather than Pakistani measurements.

**The collection rate is the softest number in the partition.** The mass balance closes by construction, but 50.3 % is an assumption, and it governs how much waste decays slowly in a landfill versus burns immediately in the street. It is a single national figure applied through an urban/rural split, when collection in practice varies between municipalities far more than between those two categories. The one Pakistani survey that asks the question directly, HIES 2024–25, reports materially less collection than the assumption (Section 6); its household-route shares are priced as a structural scenario rather than adopted, because route shares are not mass shares.

**The burned fraction is a literature value.** That 60 % of uncollected waste burns is the least evidenced parameter in the sector and has no Pakistani measurement behind it at all. It scales open burning linearly, and with it the entire air-pollutant layer in Section 8. Its complement is the fate ledger's unresolved row: 6.27 Mt of uncollected, unburned waste—dumped to land and waterways—to which no emissions pathway is applied at all. That choice is now bounded rather than silent: the scenario table prices the unresolved mass as unmanaged shallow disposal (sector total 39.87 Mt) and unmanaged deep disposal (42.28 Mt) against the reference reading (37.45 Mt), on the declared assumption that the dumped stream shares the col-

lected stream's deposit history and composition. Waterway decay lies between inert and shallow. No probability attaches to any point in the bound—a disposal-site census, not a better default, is what would close it.

**Waste composition is regional, not Pakistani.** The fossil carbon fraction governs how much of the burning $CO_2$ is reportable, and the plastics and textiles content governs the chlorine and much of the particulate. A Pakistani waste-characterisation survey would replace both.

**No methane recovery is credited and no landfill oxidation applied.** Both are set to zero because neither is documented at national scale, and both choices bias the disposal category high if any large Pakistani site does in fact manage its gas or maintain a cover. The direction of that bias is stated rather than hedged.

**The spin-up back-projection is an assumption, not data.** Extending the deposit series to 1960 at 40 % of the panel's first year is a judgement about Pakistan's waste history. Its effect is reported—12 % of current landfill methane—so a reader can remove it, but the shape of the back-projection is not evidenced.

**Uncertainties are large, and honestly so—and honestly asymmetric.** Five of the six streams carry between 74 % and 124 %; wastewater $N_2O$ carries the bounded 150-fold factor range, and the sector reads ±51.1 % by quadrature with a parameter-rerun 95% interval of 25.7 Mt to 61.4 Mt (Supplementary Note S1). These follow IPCC's own Tier 1 guidance for waste parameters and reflect that the underlying activity is modelled from population rather than measured at facilities. They are not an artefact of conservative parameter choice; they are what Tier 1 can say about a country with no facility-level waste data.

**Dioxins and furans are not estimated.** Open burning of chlorinated plastics produces them, and the hydrogen chloride reported in Section 8 is a marker for the conditions that do. Estimating them needs a dedicated study rather than a factor applied to the same tonnage.

## 11. Conclusion

Pakistan's waste sector emitted 37.45 Mt $CO_2$e in FY2023–24 on AR5 metrics: 24.46 Mt from domestic wastewater, 9.17 Mt from solid-waste disposal and 3.83 Mt from open burning. The sector is 8.4 % of a ten-sector national inventory and 87 % methane, which makes it unusually sensitive to the metric chosen and unusually relevant to near-term mitigation.

Two things generalise beyond this country.

**An activity figure shared between categories should be partitioned, not estimated twice.** Solid-waste disposal and open burning describe two fates of one waste stream, and computing them from one generation figure with $D + U = G$ asserted removes a class of inconsistency that independent estimation cannot detect. The cost is nil and the gain is that an argument about the collection rate becomes an argument about where emissions sit rather than about how much waste exists. The same shape recurs wherever two categories divide one national quantity.

**Where an assumption spans a factor approaching two and the evidence does not choose, the defensible output is both numbers.** The sewer methane correction factor moves this sector from 37.45 Mt to 28.38 Mt. The lower value happens to agree better with EDGAR, and adopting it for that reason would have been a mistake: EDGAR is a model resting on its own assumption about the same unmeasured quantity, so the agreement would have recorded a shared default as though it were evidence about Pakistani drains. We report 0.5 as central because the physical description of the drains is the stronger and more specific evidence, publish 0.1 alongside, and name what would settle it.

Domestic wastewater remains a consequential uncertainty because sewer fate and discharge conditions are not observed comprehensively. Seasonal dissolved-methane and flux evidence could distinguish the retained structural cases. Open-burning pollutant factors and burn locations require separate evidence; a population-based waste-origin allocation is not an exposure estimate. The extended release advances the existing history rather than inventing annual management observations.

## Code and data availability

The calculation engine, the activity data, the parameter tables and every output quoted here are version-controlled in the inventory repository (https://github.com/PakAirQuality/pakistan-emissions-sectors) and reproduce from a single command. The sector is regenerated with `python3 -m pkinv.sectors.waste.run_waste`; Appendix A gives the full sequence. Every number printed in this paper is read from a generated CSV rather than transcribed, and a test asserts that the manuscript and the engine agree.

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## Appendix A. Reproduction

The sector runs on Python 3.11 with no compiled dependencies:

```
pip install -e .
python3 -m pkinv.sectors.waste.run_waste
python3 -m pkinv.sectors.waste.air_pollutants
python3 -m pkinv.sectors.waste.uncertainty
```

which writes waste_totals_2024.csv, waste_by_stream_2024.csv, waste_provincial_2024.csv, waste_air_pollutants_2024.csv, waste_ledger_2024.csv, the twelve-month layer (waste_monthly_2024.csv, each month's share with the emissions it implies and the basis of the shape), the three uncertainty files (waste_uncertainty_2024.csv, waste_uncertainty_inputs_2024.csv and waste_uncertainty_scenarios_2024.csv) and waste_report.md to outputs/. The generated appendix tables are rendered by the three scripts/make_waste_*_table.py generators, and a test asserts each committed table matches a fresh regeneration. The cross-sector layer that places this sector in the national inventory is:

```
python3 -m pkinv.report.national_rollup
python3 -m pkinv.report.key_categories_all
python3 -m pkinv.report.uncertainty_all
```

Figures are regenerated with python3 -m pkinv.report.figures and copied into this manuscript by python3 scripts/sync_figures.py; a test fails if any shipped figure differs from the output it plots. The full suite, including the sector's own arithmetic tests and the checks that every number printed above matches its source CSV, runs with python3 -m unittest discover tests.

## Appendix B. Sources

Population is the 2023 Population and Housing Census (Pakistan Bureau of Statistics, 2023), interpolated to a district panel. Waste generation rates and composition derive from the Pak-EPA eight-city survey (Pakistan Environmental Protection Agency and Environmental Protection and Management Consultants, 1996), which Kaza et al. (2018) re-bins; collection coverage is the Pakistan row of Kaza et al. (2018). Sanitation shares are the WHO/UNICEF Joint Monitoring Programme's

Pakistani series (WHO/UNICEF Joint Monitoring Programme, 2023). Emission factors and model parameters are IPCC 2006 Guidelines Volume 5 defaults (IPCC, 2006), with the wastewater nitrous oxide method and its factors from the 2019 Refinement (IPCC, 2019); air-pollutant factors are the Andreae (2019) garbage-burning assessment. External comparisons are EDGAR's year-matched series (Crippa et al., 2025) and Climate TRACE (Climate TRACE, 2026).

Every factor in the repository cites its source table, and a provenance test asserts that none lacks a citation and that any informally sourced value is explicitly declared as such.

## Appendix C. Supplementary Note: the two sensitivity conventions

Both wastewater sensitivities move methane only. Nitrous oxide in 4D1 is computed from protein consumption rather than from BOD, so it is invariant under a change to either the sewer correction factor or the BOD rate.

This is worth an appendix because getting it wrong is easy and was in fact got wrong here. The sector's output file originally wrote the two sensitivity rows as *methane only* into a column whose every other row was a category total. A reader comparing the reported 24.46 Mt against the river-discharge row's 12.82 Mt would have computed a lever of 1.91 where the real one is 1.59, and the file disagreed with the module's own printed summary, which had always added the nitrous oxide back.

The rows now carry category totals and are labelled as such. The general point is that a sensitivity is only interpretable against a stated basis, and a sensitivity table mixing bases within one column is worse than no sensitivity table, because it invites a subtraction that looks valid.

## Appendix D. Supplementary Note S1: Conditional uncertainty and structural cases

The release samples the waste model's physical inputs and reruns the disposal-history, wastewater and burning calculations for each realisation. Shared waste generation, collection and uncollected fractions are not independently multiplied into category totals. The identities $D + U = G$ and $B + R = U$ are checked before a draw is accepted, where $D$ is disposed waste, $U$ uncollected waste, $G$ generated waste, $B$ burned waste and $R$ unresolved unburned waste. The latter remains an unresolved fate rather than a new emissions pathway.

Positive quantities retain their declared distributions and fractions remain bounded. The effluent $N_2O$ factor uses the existing bounded split-triangular treatment in log space over the 2019 Refinement's 0.0005–0.075 range, with 0.005 as its central mode/median. This asymmetric treatment is not replaced by a symmetric percentage around emitted $N_2O$. The 2006 method remains a separate structural case on its own factor range.

The common annual conditional ensemble uses 10 000 realisations on seed 20 261 014. For FY2023–24, the reference waste total is 37.45 Mt $CO_2e$ and its conditional 2.5th–97.5th percentiles are 25.4 Mt to 61.5 Mt on AR5. The mean exceeds the deterministic estimate because the sampled pathways are nonlinear and asymmetric. These are parameter percentiles conditional on the admitted sources and selected management interpretation, not bounds on every possible waste inventory.

The air-pollutant ensemble consumes the same burned dry mass realisations before applying pollutant-specific factor uncertainty. It does not sample population, collection or burn fraction a second time. This preserves physical dependence between open-burning GHG and pollutant emissions. Spatial and temporal allocation uncertainty remains outside these national parameter intervals.

Annual changes use paired realisations. Waste management parameters remain persistent across years unless a source explicitly supports an observed change; supported population inputs and the existing disposal state advance through the annual extension. The annual-activity-error dependence case is published separately from the persistent-parameter case. Neither is assigned a probability to combine them into one interval.

The stagnant-sewer versus flowing-discharge interpretation, burning-fraction alternatives and other supported management cases remain structural sensitivities. Their endpoints are not percentile limits and are not selected because an external inventory happens to agree more closely. Unestimated industrial wastewater and other incomplete sources remain coverage gaps rather than uncertainty draws around zero.

The earlier standalone sector ensemble remains a named diagnostic with its own seed, draw count and factor settings. It is not the current common annual conditional product. Approach 1 category screens and correlation-based driver diagnostics answer different questions: neither supplies an additive variance budget for the current nonlinear national ensemble. The most consequential unresolved evidence concerns wastewater fate and effluent release conditions, burned fractions, waste composition and actual treatment or burn locations.

## Appendix E. Supplementary Table: tier status by component

Tier status differs by component. The three categories share one activity basis – the district population panel – but carry different levels of country-specific characterisation, and two pieces of the sector are not on the IPCC tier ladder at all. One word for the whole sector would either overstate the factors or understate the activity.

## Appendix F. Supplementary Table: full factor and parameter provenance

Every parameter and factor the sector uses, with value, declared 95% uncertainty, PRICED BOUNDS where the source publishes a range, and abridged source, rendered from the versioned data files the engine reads. Note the deliberate basis differences the compact table in the body cannot carry: burning $CH_4$ is per WET tonne and $N_2O$ per DRY tonne; the $NO_x$ factor

| Component | Status | Basis |
|---|---|---|
| 4A First Order Decay | Tier 1, country-specific activity | IPCC Vol 5 Ch 3 FOD with default $k$ (0.065), $DOC_f$, $F$ and the uncategorised-site MCF (0.6) over a district census population panel and Pakistani generation rates (0.453 / 0.283 kg/cap/day, Pak-EPA 1996). The activity is country-specific; every decay parameter is a default, and the MCF stands in for a site-type census Pakistan has never published. |
| 4C2 Open burning | Tier 1, country-specific activity | IPCC Eq 5.7 over the uncollected stream (0.30 urban / 0.70 rural, What a Waste) × the Box 5.1 burned fraction (0.60), with the Pak-EPA composition supplying carbon and dry matter. The burned fraction has no Pakistani measurement behind it. |
| 4C2 air-pollutant layer | Field factors (not an IPCC tier) | Andreae 2019 garbage-burning factors on the identical burned mass. The IPCC tier ladder classifies greenhouse-gas estimation; a field-measurement factor set for co-emitted pollutants sits beside it, not on it. |
| 4D1 $CH_4$ | Tier 1, stratified activity | IPCC Vol 5 Ch 6 defaults (BOD 40 g/cap/day regional, $B_0$ 0.6) over the same population panel stratified by the JMP sanitation split - the stratification is country-specific, the factors are defaults, and the sewer MCF (0.5 central, 0.1 published alongside) is the sector's deciding judgement. |
| 4D1 $N_2O$ | Tier 1 | FAOSTAT protein through the 2019 Refinement's nitrogen chain; the effluent factor's published range spans a factor of 150, the softest number in the sector. |
| Provincial allocation | Country-specific allocation (not a tier) | Three distinct district-population-derived proxies (burning, disposal, wastewater load), reconciling exactly to each category's national figure. IPCC tiers classify estimation, not allocation. |

Table 5: Tier status by component. Generated; the parameter values quoted are asserted against `data/waste_params.csv` at generation time.

is converted to $NO_2$-equivalent (×46/30) from Andreae's published NO; and the two oxidation factors describe two different physical situations and must not be reconciled (Section 5). The bounds column exists because a published range beside a ±% that could not carry it is a contradiction: such a parameter is now either drawn from those bounds (the effluent $N_2O$ factor, bounded split-triangular in log space) or priced by named endpoint scenarios (Section Appendix D); a dash means the source states no range.

## Appendix G. Supplementary Table: the IPCC sector-4 completeness matrix

Every IPCC waste category, with its notation key (E estimated, NE not estimated, IE included elsewhere) and the justification the engine carries for every non-E entry. Generated from the same completeness table the national inventory publishes, which is itself synced bidirectionally to the gap register by test – an NE row without a register entry, or a register entry without an NE row, fails the build.

| Parameter | Value | Unit | ±% | Bounds | Source (abridged) |
|---|---|---|---|---|---|
| gen_rate_urban | 0.453 | kg/capita/day | 40 | 0.283–0.613 | Pak-EPA / EPMC 1996 National Study on Privatization of Solid Waste Management in Eight Cities of Pakistan, Table 3 (remaining urb… |
| gen_rate_rural | 0.283 | kg/capita/day | 60 | – | Pak-EPA / EPMC 1996, Table 3 (rural areas) |
| uncollected_urban | 0.3 | fraction | 30 | – | World Bank What a Waste 2.0 (Kaza et al. 2018) Pakistan row: urban collection coverage 70% |
| uncollected_rural | 0.7 | fraction | 30 | – | World Bank What a Waste 2.0 (Kaza et al. 2018) Pakistan row: rural collection coverage 30% |
| burned_fraction | 0.6 | fraction | 50 | – | IPCC 2006 GL Vol 5 Ch 5 Box 5.1 |
| oxidation_factor | 0.71 | fraction | 8 | – | IPCC 2019 Refinement Vol 5 Ch 5 Table 5.2 (Updated) Note 3; Yamada et al. 2010 |
| oxidation_factor_2006 | 0.58 | fraction | 15 | – | IPCC 2006 GL Vol 5 Ch 5 Table 5.2 Note 3 (US-EPA 1997) |
| dry_matter | 0.7 | fraction | 20 | – | Derived: Pak-EPA 1996 composition weighted by IPCC 2006 GL Vol 5 Ch 2 Table 2.4 dry-matter fractions |
| fossil_c_per_wet_t | 0.0757 | t fossil C per t wet waste | 45 | – | Derived: Pak-EPA 1996 composition x IPCC 2006 GL Vol 5 Ch 2 Table 2.4 (dm |
| total_c_per_wet_t | 0.2195 | t C per t wet waste | 35 | – | Derived: same composition and table with FCF set to 1 |
| swds_k | 0.065 | 1/yr | 25 | 0.05–0.08 | IPCC 2006 GL Vol 5 Ch 3 Table 3.3, bulk waste, tropical dry (MAT > 20 C, MAP < 1000 mm): default 0.065, range 0.05-0.08 |
| swds_delay_months | 6 | months | – | – | IPCC 2019 Refinement Vol 5 Ch 3 (reaction delay) |
| swds_docf | 0.5 | fraction | 20 | – | IPCC 2006 GL Vol 5 Ch 3 s3.2.3: 'The recommended default value for DOCf is 0.5' |
| swds_mcf | 0.6 | fraction | 35 | – | IPCC 2006 GL Vol 5 Ch 3 Table 3.1, 'Uncategorised SWDS' |
| swds_f | 0.5 | fraction | 10 | – | IPCC 2006 GL Vol 5 Ch 3 s3.2.3: 'Most waste in SWDS generates a gas with approximately 50 percent CH4 ... use of the IPCC default… |
| swds_oxidation | 0 | fraction | 0 | – | IPCC 2006 GL Vol 5 Ch 3 Table 3.2: OX = 0 for managed, unmanaged and uncategorised SWDS |
| swds_recovery_t | 0 | tonnes CH4/yr | 0 | – | No landfill-gas recovery is operating in Pakistan at national scale |
| swds_backprojection | 40 | percent | – | – | Judgement: population before 2000 is back-projected from the panel's own first-decade growth rate |
| ww_bod_g_per_cap_day | 40 | g BOD/person/day | 30 | 35–45 | IPCC 2006 GL Vol 5 Ch 6 Table 6.4, 'Asia, Middle East, Latin America' (range 35-45) |
| ww_bod_india | 34 | g BOD/person/day | 30 | 27–41 | IPCC 2006 GL Vol 5 Ch 6 Table 6.4, India (range 27-41) |
| ww_bo | 0.6 | kg CH4/kg BOD | 30 | – | IPCC 2006 GL Vol 5 Ch 6 Table 6.2, maximum CH4 producing capacity for domestic wastewater |
| ww_mcf_sewer | 0.5 | fraction | 60 | 0.4–0.8 | IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Stagnant sewer (open and warm)' (range 0.4-0.8) |
| ww_mcf_sewer_river | 0.1 | fraction | 0 | 0–0.2 | IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Sea, river and lake discharge' (range 0-0.2) |
| ww_mcf_septic | 0.5 | fraction | 0 | – | IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Septic system: half of BOD settles in anaerobic tank' (range given as 0.5, no spread) |
| ww_mcf_latrine | 0.1 | fraction | 50 | 0.05–0.15 | IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Latrine: dry climate, ground water table lower than latrine, small family (3-5 persons)' (ran… |
| ww_mcf_open | 0 | fraction | 0 | – | IPCC 2006 GL Vol 5 Ch 6 Table 6.3 has no row for open defecation |
| ww_industrial_factor | 1.25 | fraction | 20 | – | IPCC 2006 GL Vol 5 Ch 6 Eq 6.3, correction factor I for additional industrial BOD discharged into sewers (default 1.25 for collec… |
| ww_f_npr | 0.16 | kg N/kg protein | 10 | 0.15–0.17 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), Eq 6.10 and updated Table 6.11 (PDF p. 43): fra… |
| ww_f_non_con | 1.1 | fraction | 10 | 1–1.4 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), updated Table 6.11 (PDF p. 43): factor for non-… |
| ww_f_ind_com | 1.25 | fraction | 20 | 1–1.5 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), Eq 6.10 and updated Table 6.11 (PDF pp. 42-43):… |
| ww_ef_effluent | 0.005 | kg N2O-N/kg N | bounded split-tri. (log) | 0.0005–0.075 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39): freshwater, estuari… |
| ww_mcf_treated | 0.3 | fraction | 0 | 0.2–0.4 | IPCC 2006 GL Vol 5 Ch 6 Table 6.3, 'Centralised, aerobic treatment plant' - not well managed / overloaded (range 0.2-0.4; well-ma… |
| ww_treated_share | 0.01 | fraction | 0 | 0.005–0.02 | Declared scenario weight; no measured national treatment share is compiled here |
| ww_n_sludge_kg | 0 | kg N/yr | – | – | IPCC 2006 GL Vol 5 Ch 6 Eq 6.8, N_SLUDGE: nitrogen removed with sludge, default zero |
| ww_n2o_method_2019 | 1 | switch | – | – | This inventory's choice of method for domestic wastewater nitrous oxide |
| ww_f_pc | 0.96 | fraction | 5 | – | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10A (PDF p. 41): protein consumed a… |
| ww_n_hh | 1.1 | fraction | bounded split-tri. (log) | 1.07–1.17 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), Eq 6.10 (PDF p. 40): additional nitrogen from h… |
| ww_f_ind_com_decentralised | 1 | fraction | 0 | – | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), text under Table 6.10B (PDF p. 42): the industr… |
| ww_nrem_secondary | 0.4 | fraction | bounded split-tri. (log) | 0.35–0.55 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10C (PDF p. 43): nitrogen removal f… |
| ww_nrem_septic | 0.15 | fraction | bounded split-tri. (log) | 0.1–0.25 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10C (PDF p. 43): septic tank, defau… |
| ww_nrem_septic_field | 0.68 | fraction | 0 | 0.62–0.73 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10C (PDF p. 43): septic tank + land… |
| ww_nrem_latrine | 0.12 | fraction | bounded split-tri. (log) | 0.07–0.21 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.10C (PDF p. 43): latrine, default 0… |
| ww_ef_plant_aerobic | 0.016 | kg N2O-N/kg N | bounded split-tri. (log) | 0.00016–0.045 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39): centralised, aerobi… |
| ww_ef_septic | 0 | kg N2O-N/kg N | 0 | 0–0.001 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39): septic tank, 0 ('N2… |
| ww_ef_septic_field | 0.0045 | kg N2O-N/kg N | 0 | 0–0.005 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39): septic tank + land… |
| ww_ef_effluent_nutrient | 0.019 | kg N2O-N/kg N | 0 | 0.0041–0.091 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), new Table 6.8A (PDF p. 39) and Section 6.3.1.2:… |
| ww_decentralised_to_water | 1 | fraction | 0 | 0–1 | IPCC 2019 Refinement Vol 5 Ch 6 (data/raw/ipcc/19R_V5_6_Ch06_Wastewater.pdf), Eq 6.8 (PDF p. 42) against updated Figure 6.1 (… |
| ww_ef_effluent_2006_high | 0.25 | kg N2O-N/kg N | 0 | – | IPCC 2006 GL Vol 5 Ch 6 Table 6.11, emission factor for N2O from domestic wastewater effluent, upper end of the range 0.0005-0.25 |
| waste_ch4 | 6.5 | g CH4/kg wet waste | 60 | – | IPCC 2006 GL Vol 5 Ch 5 section 5.4.2 (6500 g/t wet waste; EIIP / US-EPA 2001 Vol III Ch 16 Open Burning) |
| waste_n2o | 0.15 | g N2O/kg dry waste | 100 | – | IPCC 2006 GL Vol 5 Ch 5 Table 5.6 |
| waste_gef_pm25 | 9.7 | g/kg dry matter | 45 | – | Andreae 2019 ACP Table 1 garbage burning (N=3) |
| waste_gef_co | 66 | g/kg dry matter | 50 | – | Andreae 2019 ACP Table 1 garbage burning (N=3) |
| waste_gef_nox | 2.3 | g/kg dry matter | 90 | – | Andreae 2019 ACP Table 1 garbage burning; published as NO: 1.5 (N=2) |
| waste_gef_nmvoc | 23 | g/kg dry matter | 70 | – | Andreae 2019 ACP Table 1 garbage burning total NMOGs including unidentified |
| waste_gef_bc | 1.4 | g/kg dry matter | 120 | – | Andreae 2019 ACP Table 1 garbage burning BC/EC (N=3) |
| waste_gef_so2 | 0.5 | g/kg dry matter | 80 | – | Andreae 2019 ACP Table 1 garbage burning (N=1) |
| waste_gef_nh3 | 0.9 | g/kg dry matter | 80 | – | Andreae 2019 ACP Table 1 garbage burning (N=2) |
| waste_gef_hcl | 2.8 | g/kg dry matter | 65 | – | Andreae 2019 ACP Table 1 garbage burning (N=3) |

Table 6: Full parameter and factor provenance, generated from `data/waste_params.csv` and the `waste_burning` rows of `data/emission_factors.csv`. Each repository row carries the unabridged source and a note; a provenance test asserts that no row lacks a citation and that informally sourced values are declared as such.

| Code | Category | Key | Gases | Justification (abridged) |
|---|---|---|---|---|
| 4.A | Solid waste disposal - municipal | E | CH4 | IPCC Vol 5 Ch 3 First Order Decay over a 1960-2024 deposit series, driven by the COLLECTED complement of the open-burning stream, so the two categories partition one generation figure and cannot double-count a tonne. Degradable organic carbon is the biogenic… |
| 4.A | Solid waste disposal - industrial waste | NE | CH4 | The estimated category is municipal solid waste only. Industrial solid waste reaching disposal sites has no national tonnage here; the provincial EPAs' industrial waste returns are the closing source. See waste-industrial. |
| 4.A | Solid waste disposal - construction waste | NE | CH4 | Construction and demolition waste is largely inert (near-zero degradable carbon), so its CH4 potential is small by composition - but its tonnage is unquantified and it is a distinct absence from industrial waste, with a different closing source: the Urban Uni… |
| 4.A | Solid waste disposal - CO2 from landfill gas | IE | CO2 | Biogenic, and unlike open burning IPCC does not request it as a memo item: that carbon is tracked as a stock change in AFOLU. |
| 4.B | Biological treatment of solid waste (composting, anaerobic… | NE | CH4/N2O | Municipal composting operates at city scale - Lahore's composting facility is the largest, with smaller plants elsewhere - but no national throughput series has been compiled here. Believed small against 4.A, and the omission biases 4.A UP rather than the tot… |
| 4.C.1 | Waste incineration | NE | CO2/CH4/N2O | No municipal-scale incineration operates in Pakistan, but hospital and clinical-waste incinerators do; their throughput has not been obtained here, so no magnitude is stated. See waste-incineration. |
| 4.C.2 | Open burning of waste | E | CO2/CH4/N2O | IPCC Eq 5.7 over the UNCOLLECTED stream x a 0.60 burned fraction. Only fossil CO2 counts; biogenic CO2 is a memo item. |
| 4.D.1 | Domestic wastewater | E | CH4/N2O | IPCC Vol 5 Ch 6 on the district population panel, stratified by the WHO/UNICEF JMP sanitation split - sewer 62.6% urban against 8.8% rural, which is why the stratification matters. THE JUDGEMENT THAT DECIDES THIS CATEGORY is the sewer MCF: IPCC gives stagna… |
| 4.D | Industrial wastewater | NE | CH4 | Needs industrial output by sector and the organic load of each sector's wastewater; neither is held. |

Table 7: The waste-sector completeness matrix, generated from the engine (`outputs/completeness.csv`, sector-4 rows). Full justifications are in the generated national table; each NE row names its gap-register entry.